Paper/Subject	B.E/B.Tech.(Paper 1)
Exam Date	26 Jul 2022
Exam Slot	1

Question ID:100101
Topic Name: Mathematics-Section A
Question:

Let $f: \mathbf{R} \to \mathbf{R}$ be a continuous function such that $f(3x) - f(x) = x$. If $f(8) = 7$, then $f(14)$ is equal to :

A 4

B 10

C 11

D 16

Answer Given By Candidate:**Not Attempted**

Question ID:100102
Topic Name: Mathematics-Section A
Question:

Let O be the origin and A be the point $z_1 = 1 + 2i$. If B is the point z_2, $\mathrm{Re}(z_2) < 0$, such that OAB is a right angled isosceles triangle with OB as hypotenuse, then which of the following is **NOT** true ?

A $\arg z_2 = \pi - \tan^{-1} 3$

B $\arg (z_1 - 2z_2) = -\tan^{-1}\frac{4}{3}$

C $|z_2| = \sqrt{10}$

D $|2z_1 - z_2| = 5$

Answer Given By Candidate:**D**

Question ID:100103
Topic Name: Mathematics-Section A

Question:

If the system of linear equations.

$$8x+y+4z=-2$$
$$x+y+z=0$$
$$\lambda x-3y=\mu$$

has infinitely many solutions, then the distance of the point $\left(\lambda, \mu, -\frac{1}{2}\right)$ from the plane $8x+y+4z+2=0$ is :

A $3\sqrt{5}$

B 4

C $\frac{26}{9}$

D $\frac{10}{3}$

Answer Given By Candidate:**Not Attempted**

Question ID:100104
Topic Name: Mathematics-Section A
Question:

Let A be a 2×2 matrix with $\det(A)=-1$ and $\det((A+I)(\text{Adj}(A)+I))=4$. Then the sum of the diagonal elements of A can be :

A -1

B 2

C 1

D $-\sqrt{2}$

Answer Given By Candidate:**D**

Question ID:100105
Topic Name: Mathematics-Section A
Question:

The odd natural number a, such that the area of the region bounded by $y=1$, $y=3$, $x=0$, $x=y^a$ is $\frac{364}{3}$, is equal to :

A 3

B 5

C 7

D 9

Answer Given By Candidate:**Not Attempted**

Question ID:100106
Topic Name: Mathematics-Section A
Question:

Consider two G.Ps. 2, 2^2, 2^3, ... and 4, 4^2, 4^3, ... of 60 and n terms respectively. If the geometric mean of all the $60+n$ terms is $(2)^{\frac{225}{8}}$, then $\sum_{k=1}^{n} k(n-k)$ is equal to :

A 560

B 1540

C 1330

D 2600

Answer Given By Candidate:**Not Attempted**

Question ID:100107
Topic Name: Mathematics-Section A
Question:

If the function $f(x) = \begin{cases} \frac{\log_e(1-x+x^2)+\log_e(1+x+x^2)}{\sec x-\cos x} & , \quad x \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right) - \{0\} \\ k & , \quad x = 0 \end{cases}$

is continuous at $x=0$, then k is equal to :

A 1

B -1

C e

D 0

Answer Given By Candidate:**Not Attempted**

Question ID:100108
Topic Name: Mathematics-Section A
Question:

If $f(x) = \begin{cases} x+a & , \quad x \le 0 \\ |x-4| & , \quad x > 0 \end{cases}$ and $g(x) = \begin{cases} x+1 & , \quad x < 0 \\ (x-4)^2 + b & , \quad x \ge 0 \end{cases}$

are continuous on **R**, then $(gof)(2)+(fog)(-2)$ is equal to :

A -10

B 10

C 8 D

Answer Given By Candidate: Not Attempted

Question ID: 100109
Topic Name: Mathematics-Section A

Question: Let $f(x) = \begin{cases} x^3 - x^2 + 10x - 7, & x \le 1 \\ -2x + \log_2 (b^2 - 4), & x > 1 \end{cases}$. Then the set of all values of b, for which $f(x)$ has maximum value at $x = 1$, is :

A $(-6, -2)$

B $(2, 6)$

C $[-6, -2) \cup (2, 6]$

D $\left[-\sqrt{6}, -2\right) \cup \left(2, \sqrt{6}\right]$

Answer Given By Candidate: Not Attempted

Question ID: 100110
Topic Name: Mathematics-Section A

Question: If $a = \lim\limits_{n\to\infty} \sum\limits_{k=1}^{n} \frac{2n}{n^2 + k^2}$ and $f(x) = \sqrt{\frac{1 - \cos x}{1 + \cos x}}$, $x \in (0, 1)$, then :

A $2\sqrt{2} f\left(\frac{a}{2}\right) = f'\left(\frac{a}{2}\right)$

B $f\left(\frac{a}{2}\right) f'\left(\frac{a}{2}\right) = \sqrt{2}$

C $\sqrt{2} f\left(\frac{a}{2}\right) = f'\left(\frac{a}{2}\right)$

D $f\left(\frac{a}{2}\right) = \sqrt{2} f'\left(\frac{a}{2}\right)$

Answer Given By Candidate: Not Attempted

Question ID: 100111
Topic Name: Mathematics-Section A

Question:

If $\frac{dy}{dx} + 2y\tan x = \sin x, 0 < x < \frac{\pi}{2}$ and $y\left(\frac{\pi}{3}\right) = 0$, then the maximum value of $y(x)$ is :

A $\frac{1}{8}$

B $\frac{3}{4}$

C $\frac{1}{4}$

D $\frac{3}{8}$

Answer Given By Candidate:**A**

Question ID:100112
Topic Name: Mathematics-Section A
Question:

A point P moves so that the sum of squares of its distances from the points (1, 2) and $(-2, 1)$ is 14. Let $f(x, y) = 0$ be the locus of P, which intersects the x-axis at the points A, B and the y-axis at the points C, D. Then the area of the quadrilateral ACBD is equal to :

A $\frac{9}{2}$

B $\frac{3\sqrt{17}}{2}$

C $\frac{3\sqrt{17}}{4}$

D 9

Answer Given By Candidate:**Not Attempted**

Question ID:100113
Topic Name: Mathematics-Section A
Question:

Let the tangent drawn to the parabola $y^2 = 24x$ at the point (α, β) is perpendicular to the line $2x + 2y = 5$. Then the normal to the hyperbola $\frac{x^2}{\alpha^2} - \frac{y^2}{\beta^2} = 1$ at the point $(\alpha + 4, \beta + 4)$ does **NOT** pass through the point :

A $(25, 10)$

B $(20, 12)$

C $(30, 8)$ D $(15, 13)$

Answer Given By Candidate:**D**

Question ID:100114
Topic Name: Mathematics-Section A
Question:

The length of the perpendicular from the point $(1, -2, 5)$ on the line passing through $(1, 2, 4)$ and parallel to the line $x+y-z=0=x-2y+3z-5$ is :

A $\sqrt{\frac{21}{2}}$

B $\sqrt{\frac{9}{2}}$

C $\sqrt{\frac{73}{2}}$

D 1

Answer Given By Candidate:**Not Attempted**

Question ID:100115
Topic Name: Mathematics-Section A
Question:

Let $\vec{a} = \alpha\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - \alpha\hat{k}$, $\alpha > 0$. If the projection of $\vec{a} \times \vec{b}$ on the vector $-\hat{i} + 2\hat{j} - 2\hat{k}$ is 30, then α is equal to :

A $\frac{15}{2}$

B 8

C $\frac{13}{2}$

D 7

Answer Given By Candidate:**D**

Question ID:100116
Topic Name: Mathematics-Section A
Question:

The mean and variance of a binomial distribution are α and $\frac{\alpha}{3}$ respectively.

If $P(X=1)=\frac{4}{243}$, then $P(X=4 \text{ or } 5)$ is equal to :

A $\frac{5}{9}$

B $\frac{64}{81}$

C $\frac{16}{27}$

D $\frac{145}{243}$

Answer Given By Candidate:**Not Attempted**

Question ID:100117
Topic Name: Mathematics-Section A
Question:

Let E_1, E_2, E_3 be three mutually exclusive events such that $P(E_1)=\frac{2+3p}{6}$, $P(E_2)=\frac{2-p}{8}$ and $P(E_3)=\frac{1-p}{2}$. If the maximum and minimum values of p are p_1 and p_2, then (p_1+p_2) is equal to :

A $\frac{2}{3}$

B $\frac{5}{3}$

C $\frac{5}{4}$

D 1

Answer Given By Candidate:**Not Attempted**

Question ID:100118

Topic Name: Mathematics-Section A
Question:

Let $S=\{\theta\epsilon\ [0,\ 2\pi] : 8^{2\sin^2\theta}+8^{2\cos^2\theta}=16\}$. Then $n(S)+\sum_{\theta\epsilon S}\left(\sec\left(\frac{\pi}{4}+2\theta\right)\operatorname{cosec}\left(\frac{\pi}{4}+2\theta\right)\right)$ is equal to :

A 0

B -2

C -4

D 12

Answer Given By Candidate:**Not Attempted**

Question ID:100119
Topic Name: Mathematics-Section A
Question: $\tan\left(2\tan^{-1}\frac{1}{5}+\sec^{-1}\frac{\sqrt{5}}{2}+2\tan^{-1}\frac{1}{8}\right)$ is equal to :

A 1

B 2

C $\frac{1}{4}$

D $\frac{5}{4}$

Answer Given By Candidate:**B**

Question ID:100120
Topic Name: Mathematics-Section A
Question: The statement $(\sim(p \Leftrightarrow \sim q)) \wedge q$ is :

A a tautology

B a contradiction

C equivalent to $(p \Rightarrow q) \wedge q$

D equivalent to $(p \Rightarrow q) \wedge p$

Answer Given By Candidate:**Not Attempted**

Question ID:100121
Topic Name: Mathematics-Section B

Question:

If for some p, q, r $\in$ **R**, not all have same sign, one of the roots of the equation $(p^2+q^2)x^2-2q(p+r)x+q^2+r^2=0$ is also a root of the equation $x^2+2x-8=0$, then $\frac{q^2+r^2}{p^2}$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100122
Topic Name: Mathematics-Section B
Question:

The number of 5-digit natural numbers, such that the product of their digits is 36, is _____.

Answer Given By Candidate:**36**

Question ID:100123
Topic Name: Mathematics-Section B
Question:

The series of positive multiples of 3 is divided into sets : {3}, {6, 9, 12}, {15, 18, 21, 24, 27},. . . Then the sum of the elements in the 11th set is equal to _____.

Answer Given By Candidate:**3587**

Question ID:100124
Topic Name: Mathematics-Section B
Question:

The number of distinct real roots of the equation $x^5(x^3-x^2-x+1)+x\ (3x^3-4x^2-2x+4)-1=0$ is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100125
Topic Name: Mathematics-Section B
Question:

If the coefficients of x and x^2 in the expansion of $(1+x)^p\ (1-x)^q$, p, q $\leq$ 15, are -3 and -5 respectively, then the coefficient of x^3 is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100126
Topic Name: Mathematics-Section B
Question:

If $n(2n+1)\int_0^1(1-x^n)^{2n}\,dx = 1177\int_0^1(1-x^n)^{2n+1}\,dx$, then n$\in$**N** is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100127
Topic Name: Mathematics-Section B

Question:

Let a curve $y=y(x)$ pass through the point (3, 3) and the area of the region under this curve, above the x-axis and between the abscissae 3 and $x(>3)$ be $\left(\frac{y}{x}\right)^3$. If this curve also passes through the point $(\alpha, 6\sqrt{10})$ in the first quadrant, then α is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100128
Topic Name: Mathematics-Section B
Question:

The equations of the sides AB, BC and CA of a triangle ABC are $2x+y=0$, $x+py=15a$ and $x-y=3$ respectively. If its orthocentre is (2, a), $-\frac{1}{2} < a < 2$, then p is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100129
Topic Name: Mathematics-Section B
Question:

Let the function $f(x)=2x^2-\log_e x$, $x > 0$, be decreasing in (0, a) and increasing in (a, 4). A tangent to the parabola $y^2=4ax$ at a point P on it passes through the point (8a, 8a − 1) but does not pass through the point $\left(-\frac{1}{a}, 0\right)$. If the equation of the normal at P is $\frac{x}{\alpha}+\frac{y}{\beta}=1$, then $\alpha+\beta$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100130
Topic Name: Mathematics-Section B
Question:

Let Q and R be two points on the line $\frac{x+1}{2}=\frac{y+2}{3}=\frac{z-1}{2}$ at a distance $\sqrt{26}$ from the point P(4, 2, 7). Then the square of the area of the triangle PQR is _____.

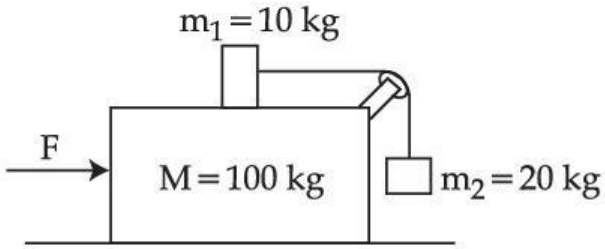
Answer Given By Candidate:**Not Attempted**

Question ID:100131
Topic Name: Physics-Section A

Question:

Three masses $M = 100$ kg, $m_1 = 10$ kg and $m_2 = 20$ kg are arranged in a system as shown in figure. All the surfaces are frictionless and strings are inextensible and weightless. The pulleys are also weightless and frictionless. A force F is applied on the system so that the mass m_2 moves upward with an acceleration of 2 ms^{-2}. The value of F is :

(Take $g = 10\ ms^{-2}$)



A 3360 N

B 3380 N

C 3120 N

D 3240 N

Answer Given By Candidate: **Not Attempted**

Question ID: 100132
Topic Name: Physics-Section A
Question:

A radio can tune to any station in 6 MHz to 10 MHz band. The value of corresponding wavelength bandwidth will be :

A 4 m

B 20 m

C 30 m

D 50 m

Answer Given By Candidate: **Not Attempted**

Question ID: 100133
Topic Name: Physics-Section A
Question:

The disintegration rate of a certain radioactive sample at any instant is 4250 disintegrations per minute. 10 minutes later, the rate becomes 2250 disintegrations per minute. The approximate decay constant is :

(Take $\log_{10} 1.88 = 0.274$)

A

$0.02\ \text{min}^{-1}$

B $2.7\ \text{min}^{-1}$

C $0.063\ \text{min}^{-1}$

D $6.3\ \text{min}^{-1}$

Answer Given By Candidate: **Not Attempted**

Question ID: 100134
Topic Name: Physics-Section A
Question:

A parallel beam of light of wavelength 900 nm and intensity 100 Wm^{-2} is incident on a surface perpendicular to the beam. The number of photons crossing 1 cm^2 area perpendicular to the beam in one second is :

A 3×10^{16}

B 4.5×10^{16}

C 4.5×10^{17}

D 4.5×10^{20}

Answer Given By Candidate: **Not Attempted**

Question ID: 100135
Topic Name: Physics-Section A
Question:

In Young's double slit experiment, the fringe width is 12 mm. If the entire arrangement is placed in water of refractive index $\frac{4}{3}$, then the fringe width becomes (in mm) :

A 16

B 9

C 48

D 12

Answer Given By Candidate: **A**

Question ID: 100136
Topic Name: Physics-Section A

The magnetic field of a plane electromagnetic wave is given by :

$\vec{B} = 2 \times 10^{-8} \sin(0.5 \times 10^{3} x + 1.5 \times 10^{11} t)\ \hat{j}$ T.

Question: The amplitude of the electric field would be :

A $6\ \mathrm{Vm}^{-1}$ along x-axis

B $3\ \mathrm{Vm}^{-1}$ along z-axis

C $6\ \mathrm{Vm}^{-1}$ along z-axis

D $2 \times 10^{-8}\ \mathrm{Vm}^{-1}$ along z-axis

Answer Given By Candidate:**C**

Question ID:100137
Topic Name: Physics-Section A
Question:

In a series LR circuit $X_L = R$ and power factor of the circuit is P_1. When capacitor with capacitance C such that $X_L = X_C$ is put in series, the power factor becomes P_2. The ratio $\frac{P_1}{P_2}$ is :

A $\frac{1}{2}$

B $\frac{1}{\sqrt{2}}$

C $\frac{\sqrt{3}}{\sqrt{2}}$

D $2 : 1$

Answer Given By Candidate:**Not Attempted**

Question ID:100138
Topic Name: Physics-Section A
Question:

A charge particle is moving in a uniform magnetic field $\left(2\hat{i} + 3\hat{j}\right)$ T. If it has an acceleration of $\left(\alpha\hat{i} - 4\hat{j}\right)$ m/s^2, then the value of α will be :

A 3

B 6

C 12

D 2

Answer Given By Candidate:**Not Attempted**

Question ID:100139
Topic Name: Physics-Section A
Question:

B_X and B_Y are the magnetic fields at the centre of two coils X and Y respectively each carrying equal current. If coil X has 200 turns and 20 cm radius and coil Y has 400 turns and 20 cm radius, the ratio of B_X and B_Y is :

A 1 : 1

B 1 : 2

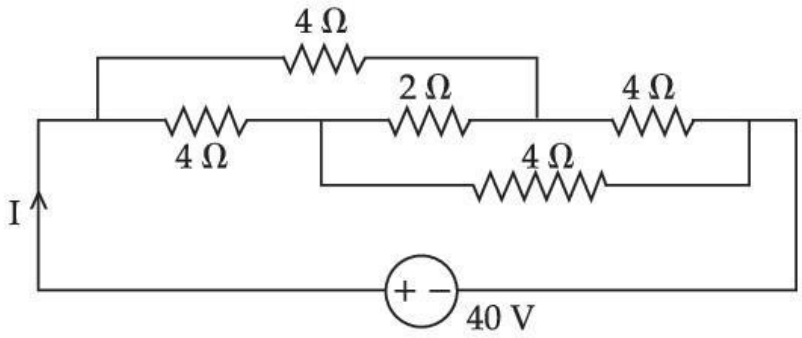
C 2 : 1

D 4 : 1

Answer Given By Candidate:**Not Attempted**

Question ID:100140
Topic Name: Physics-Section A

The current I in the given circuit will be :



Question:

A 10 A

B 20 A

C 4 A

D 40 A

Answer Given By Candidate:**Not Attempted**

Question ID:100141
Topic Name: Physics-Section A

Question:

The total charge on the system of capacitors $C_1 = 1\ \mu F$, $C_2 = 2\ \mu F$, $C_3 = 4\ \mu F$ and $C_4 = 3\ \mu F$ connected in parallel is :

(Assume a battery of 20 V is connected to the combination)

A $200\ \mu C$

B $200\ C$

C $10\ \mu C$

D $10\ C$

Answer Given By Candidate: **Not Attempted**

Question ID: 100142
Topic Name: Physics-Section A
Question:

When a particle executes Simple Hormonic Motion, the nature of graph of velocity as a function of displacement will be :

A Circular

B Elliptical

C Sinusoidal

D Straight line

Answer Given By Candidate: **Not Attempted**

Question ID: 100143
Topic Name: Physics-Section A
Question:

7 mol of a certain monoatomic ideal gas undergoes a temperature increase of 40K at constant pressure. The increase in the internal energy of the gas in this process is :

(Given $R = 8.3\ JK^{-1}\ mol^{-1}$)

A 5810 J

B 3486 J

C 11620 J

D 6972 J

Answer Given By Candidate: **Not Attempted**

Question ID: 100144

Topic Name: Physics-Section A
Question:

A monoatomic gas at pressure P and volume V is suddenly compressed to one eighth of its original volume. The final pressure at constant entropy will be :

A P

B 8P

C 32P

D 64P

Answer Given By Candidate:**Not Attempted**

Question ID:100145
Topic Name: Physics-Section A
Question:

A water drop of radius 1 cm is broken into 729 equal droplets. If surface tension of water is 75 dyne/cm, then the gain in surface energy upto first decimal place will be :

(Given $\pi = 3.14$)

A 8.5×10^{-4} J

B 8.2×10^{-4} J

C 7.5×10^{-4} J

D 5.3×10^{-4} J

Answer Given By Candidate:**Not Attempted**

Question ID:100146
Topic Name: Physics-Section A
Question:

The percentage decrease in the weight of a rocket, when taken to a height of 32 km above the surface of earth will, be :

(Radius of earth = 6400 km)

A 1%

B 3%

C 4%

D 0.5%

Answer Given By Candidate:**Not Attempted**

Question ID:100147
Topic Name: Physics-Section A
Question:

As per the given figure, two blocks each of mass 250 g are connected to a spring of spring constant $2\,\mathrm{Nm}^{-1}$. If both are given velocity v in opposite directions, then maximum elongation of the spring is :



A $\frac{v}{2\sqrt{2}}$

B $\frac{v}{2}$

C $\frac{v}{4}$

D $\frac{v}{\sqrt{2}}$

Answer Given By Candidate:**Not Attempted**

Question ID:100148
Topic Name: Physics-Section A
Question:

A monkey of mass 50 kg climbs on a rope which can withstand the tension (T) of 350 N. If monkey initially climbs down with an acceleration of 4 $\mathrm{m/s^2}$ and then climbs up with an acceleration of 5 $\mathrm{m/s^2}$. Choose the correct option ($g = 10\ \mathrm{m/s^2}$).

A T = 700 N while climbing upward

B T = 350 N while going downward

C Rope will break while climbing upward

D Rope will break while going downward

Answer Given By Candidate:**Not Attempted**

Question ID:100149
Topic Name: Physics-Section A
Question:

Two projectiles thrown at 30° and 45° with the horizontal respectively, reach the maximum height in same time. The ratio of their initial velocities is :

A

$1:\sqrt{2}$

B $2:1$

C $\sqrt{2}:1$

D $1:2$

Answer Given By Candidate:**Not Attempted**

Question ID:100150
Topic Name: Physics-Section A
Question:

A screw gauge of pitch 0.5 mm is used to measure the diameter of uniform wire of length 6.8 cm, the main scale reading is 1.5 mm and circular scale reading is 7. The calculated curved surface area of wire to appropriate significant figures is :

[Screw gauge has 50 divisions on its circular scale]

A $6.8\ cm^2$

B $3.4\ cm^2$

C $3.9\ cm^2$

D $2.4\ cm^2$

Answer Given By Candidate:**Not Attempted**

Question ID:100151
Topic Name: Physics-Section B
Question:

If the initial velocity in horizontal direction of a projectile is unit vector $\hat{i}$ and the equation of trajectory is $y=5x(1-x)$. The y component vector of the initial velocity is _____ $\hat{j}$.

(Take $g=10\ m/s^2$)

Answer Given By Candidate:**Not Attempted**

Question ID:100152
Topic Name: Physics-Section B

Question:

A disc of mass 1 kg and radius R is free to rotate about a horizontal axis passing through its centre and perpendicular to the plane of disc. A body of same mass as that of disc is fixed at the highest point of the disc. Now the system is released, when the body comes to the lowest position, its angular speed will be $4\sqrt{\frac{x}{3R}}$ rad s^{-1} where $x =$ _____.

$(g = 10\ ms^{-2})$

Answer Given By Candidate:**Not Attempted**

Question ID:100153
Topic Name: Physics-Section B
Question:

In an experiment to determine the Young's modulus of wire of a length exactly 1 m, the extension in the length of the wire is measured as 0.4 mm with an uncertainty of ± 0.02 mm when a load of 1 kg is applied. The diameter of the wire is measured as 0.4 mm with an uncertainty of ± 0.01 mm. The error in the measurement of Young's modulus (ΔY) is found to be $x \times 10^{10}$ Nm^{-2}. The value of x is _____.

(take $g = 10\ ms^{-2}$)

Answer Given By Candidate:**Not Attempted**

Question ID:100154
Topic Name: Physics-Section B
Question:

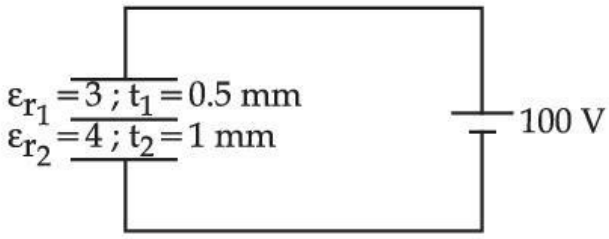
When a car is approaching the observer, the frequency of horn is 100 Hz. After passing the observer, it is 50 Hz. If the observer moves with the car, the frequency will be $\frac{x}{3}$ Hz where $x =$ _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100155
Topic Name: Physics-Section B

Question:

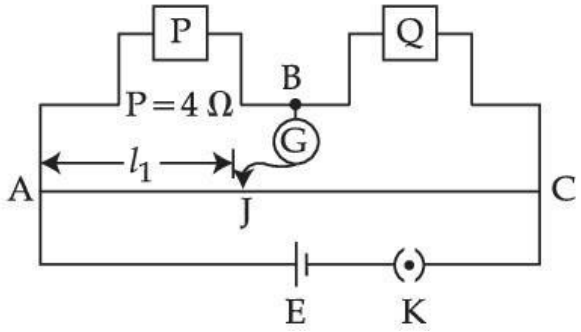
A composite parallel plate capacitor is made up of two different dielectric materials with different thickness (t_1 and t_2) as shown in figure. The two different dielectric materials are separated by a conducting foil F. The voltage of the conducting foil is _____ V.



Answer Given By Candidate:**Not Attempted**

Question ID:100156
Topic Name: Physics-Section B
Question:

Resistances are connected in a meter bridge circuit as shown in the figure. The balancing length l_1 is 40 cm. Now an unknown resistance x is connected in series with P and new balancing length is found to be 80 cm measured from the same end. Then the value of x will be _____ Ω.

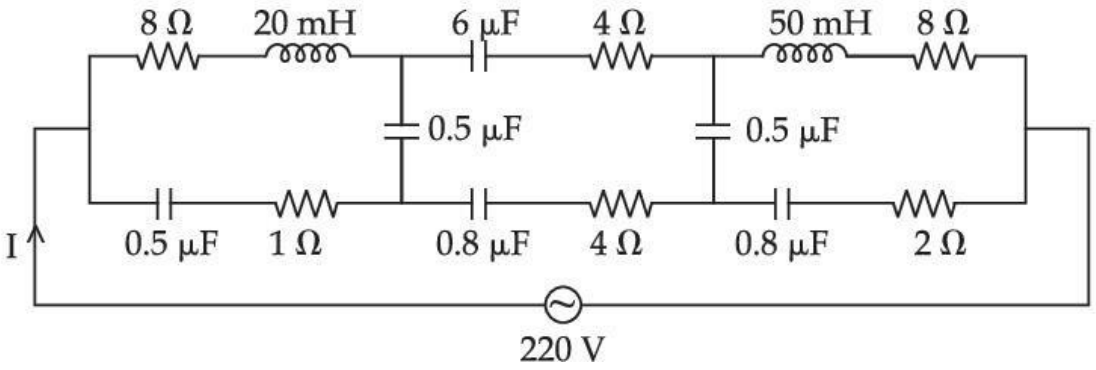


Answer Given By Candidate:**Not Attempted**

Question ID:100157
Topic Name: Physics-Section B

Question:

The effective current I in the given circuit at very high frequencies will be _____ A.

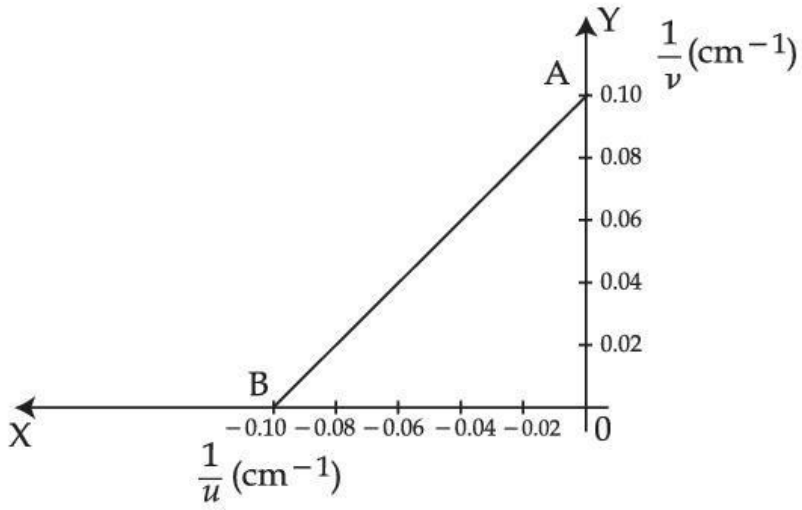


Answer Given By Candidate:**Not Attempted**

Question ID:100158
Topic Name: Physics-Section B
Question:

The graph between $\frac{1}{u}$ and $\frac{1}{v}$ for a thin convex lens in order to determine its focal length is plotted as shown in the figure. The refractive index of lens is 1.5 and its both the surfaces have same radius of curvature R. The value of R will be _____ cm.

(where u = object distance, v = image distance)



Answer Given By Candidate:**2**

Question ID:100159
Topic Name: Physics-Section B

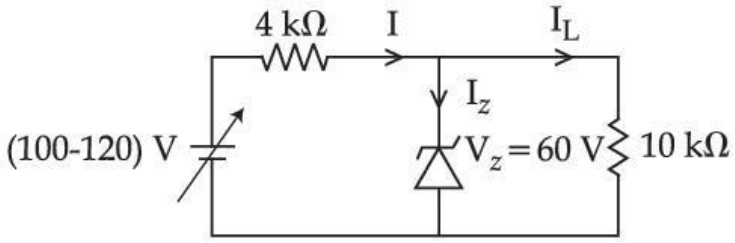
Question:

In the hydrogen spectrum, λ be the wavelength of first transition line of Lyman series. The wavelength difference will be "$a\lambda$" between the wavelength of 3rd transition line of Paschen series and that of 2nd transition line of Balmer series where a = _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100160
Topic Name: Physics-Section B
Question:

In the circuit shown below, maximum zener diode current will be _____ mA.



Answer Given By Candidate:**Not Attempted**

Question ID:100161
Topic Name: Chemistry-Section A

Match **List - I** with **List - II**.

	List - I (Compound)		**List - II (Shape)**
(A)	BrF_5	(I)	bent
(B)	$[CrF_6]^{3-}$	(II)	square pyramidal
(C)	O_3	(III)	trigonal bipyramidal
(D)	PCl_5	(IV)	octahedral

Question:

Choose the **correct** answer from the options given below :

A (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

B (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

C (A) - (II), (B) - (IV), (C) - (I), (D) - (III)

D (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

Answer Given By Candidate:**C**

Question ID:100162
Topic Name: Chemistry-Section A

Match **List - I** with **List - II**.

	List - I (Processes/Reactions)		**List - II (Catalyst)**
(A)	$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$	(I)	Fe(s)
(B)	$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$	(II)	$Pt(s) - Rh(s)$
(C)	$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$	(III)	V_2O_5
(D)	Vegetable oil(l) + H_2 → Vegetable ghee(s)	(IV)	Ni(s)

Question: Choose the **correct** answer from the options given below :

A (A) - (III), (B) - (I), (C) - (II), (D) - (IV)

B (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

C (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

D (A) - (IV), (B) - (II), (C) - (III), (D) - (I)

Answer Given By Candidate: **D**

Question ID: 100163
Topic Name: Chemistry-Section A
Question:

Given two statements below :

Statement I : In Cl_2 molecule the covalent radius is double of the atomic radius of chlorine.

Statement II : Radius of anionic species is always greater than their parent atomic radius.

Choose the **most appropriate** answer from options given below :

A Both **Statement I** and **Statement II** are correct.

B Both **Statement I** and **Statement II** are incorrect.

C **Statement I** is correct but **Statement II** is incorrect.

D **Statement I** is incorrect but **Statement II** is correct.

Answer Given By Candidate: **Not Attempted**

Question ID: 100164
Topic Name: Chemistry-Section A
Question: Refining using liquation method is the most suitable for metals with :

A Low melting point

B High boiling point

C High electrical conductivity

D Less tendency to be soluble in melts than impurities

Answer Given By Candidate: **Not Attempted**

Question ID: 100165
Topic Name: Chemistry-Section A
Question: Which of the following can be used to prevent the decomposition of H_2O_2 ?

A Urea

B Formaldehyde

C Formic acid

D Ethanol

Answer Given By Candidate: **Not Attempted**

Question ID: 100166
Topic Name: Chemistry-Section A
Question: Reaction of $BeCl_2$ with $LiAlH_4$ gives :

(A) $AlCl_3$

(B) BeH_2

(C) LiH

(D) LiCl

(E) $BeAlH_4$

Choose the **correct** answer from options given below :

A (A), (D) and (E)

B (A), (B) and (D)

C (D) and (E)

D (B), (C) and (D)

Answer Given By Candidate: **B**

Question ID: 100167
Topic Name: Chemistry-Section A
Question: Borazine, also known as inorganic benzene, can be prepared by the reaction of 3-equivalents of "X" with 6-equivalents of "Y". "X" and "Y", respectively are :

A $B(OH)_3$ and NH_3

B B_2H_6 and NH_3

C B_2H_6 and HN_3

D NH_3 and B_2O_3

Answer Given By Candidate: **Not Attempted**

Question ID: 100168
Topic Name: Chemistry-Section A
Question: Which of the given reactions is not an example of disproportionation reaction ?

A $2H_2O_2 \rightarrow 2H_2O + O_2$

B $2NO_2 + H_2O \rightarrow HNO_3 + HNO_2$

C $MnO_4^- + 4H^+ + 3e^- \rightarrow MnO_2 + 2H_2O$

D $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

Answer Given By Candidate: **Not Attempted**

Question ID: 100169
Topic Name: Chemistry-Section A
Question:
The dark purple colour of $KMnO_4$ disappears in the titration with oxalic acid in acidic medium. The overall change in the oxidation number of manganese in the reaction is :

A 5

B 1

C 7

D 2

Answer Given By Candidate: **Not Attempted**

Question ID: 100170
Topic Name: Chemistry-Section A

$$\dot{C}l + CH_4 \rightarrow A + B$$

Question: A and B in the above atmospheric reaction step are :

A C_2H_6 and Cl_2

B

$\dot{C}HCl_2$ and H_2

C $\dot{C}H_3$ and HCl

D C_2H_6 and HCl

Answer Given By Candidate:C

Question ID:100171
Topic Name: Chemistry-Section A
Question:
Which technique among the following, is most appropriate in separation of a mixture of 100 mg of *p*-nitrophenol and picric acid ?

A Steam distillation

B 2-5 ft long column of silica gel

C Sublimation

D Preparative TLC (Thin Layer Chromatography)

Answer Given By Candidate:Not Attempted

Question ID:100172
Topic Name: Chemistry-Section A
Question:
The difference in the reaction of phenol with bromine in chloroform and bromine in water medium is due to :

A Hyperconjugation in substrate

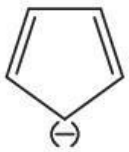
B Polarity of solvent

C Free radical formation

D Electromeric effect the substrate

Answer Given By Candidate:Not Attempted

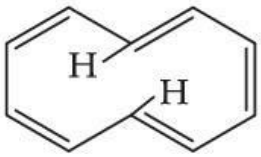
Question ID:100173
Topic Name: Chemistry-Section A
Question: Which of the following compounds is **not** aromatic ?

A 

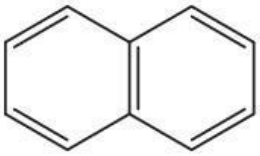
B



C



D

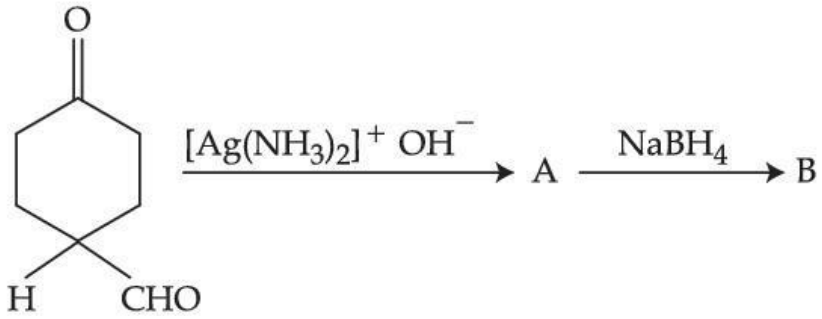


Answer Given By Candidate: **Not Attempted**

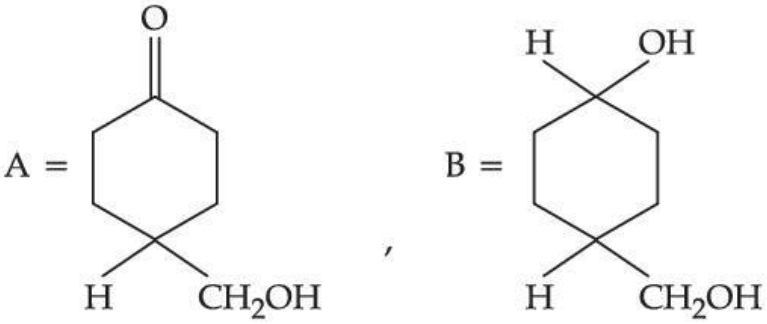
Question ID: 100174

Topic Name: Chemistry-Section A

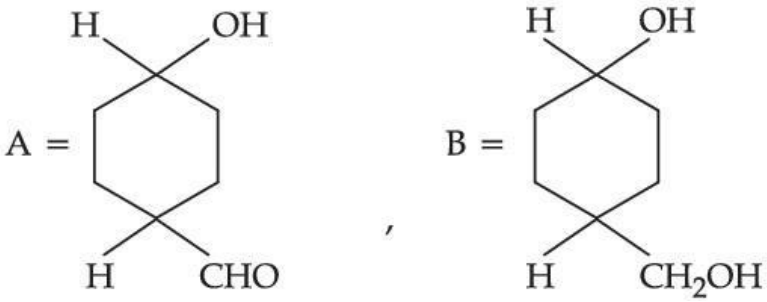
Question: The products formed in the following reaction, **A** and **B** are



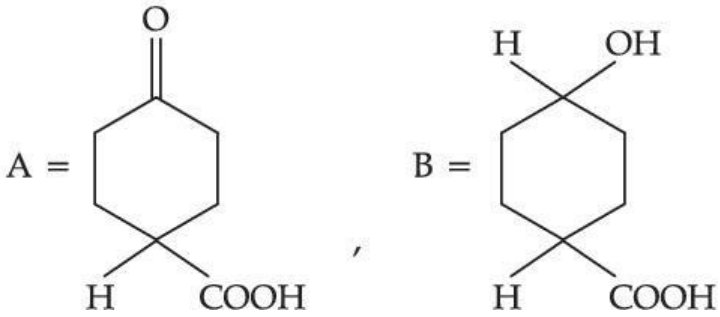
A



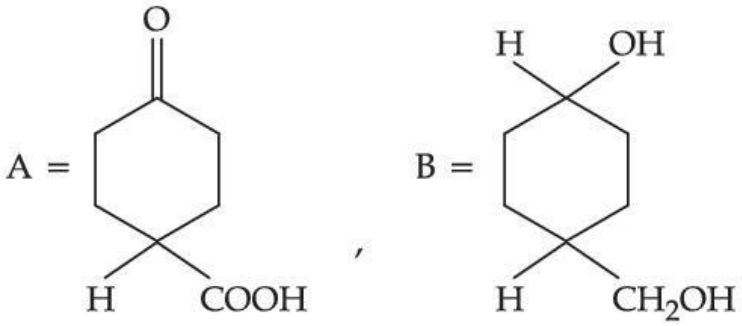
B



C



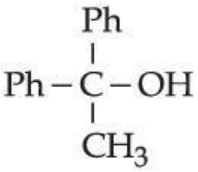
D



Answer Given By Candidate: **Not Attempted**

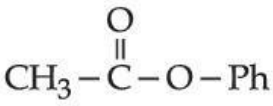
Question ID: 100175
Topic Name: Chemistry-Section A
Question:

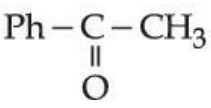
Which reactant will give the following alcohol on reaction with one mole of phenyl magnesium bromide (PhMgBr) followed by acidic hydrolysis ?



A $CH_3-C\equiv N$

B $Ph-C\equiv N$

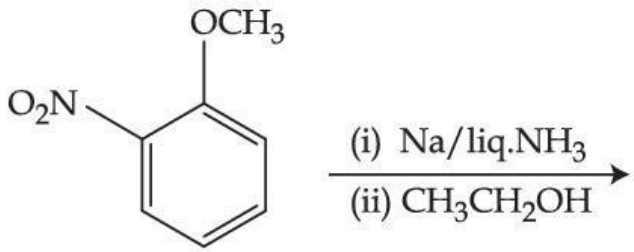
C 

D 

Answer Given By Candidate: **C**

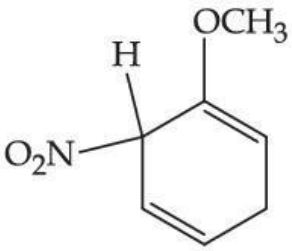
Question ID: 100176
Topic Name: Chemistry-Section A

The major product of the following reaction is

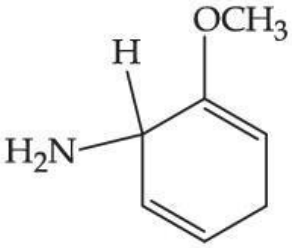


Question:

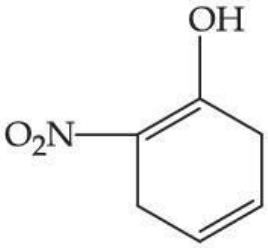
A



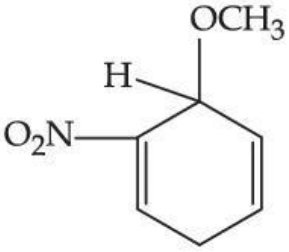
B



C



D

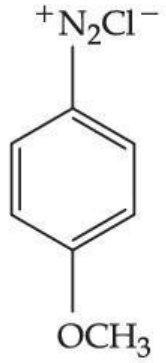


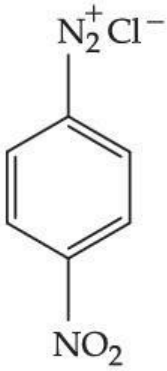
Answer Given By Candidate:C

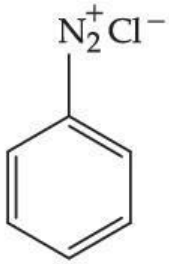
Question ID:100177
Topic Name: Chemistry-Section A

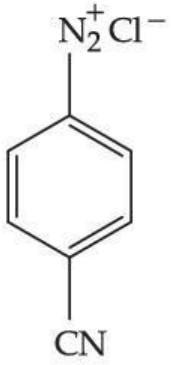
Question:

The correct stability order of the following diazonium salt is

(A) 

(B) 

(C) 

(D) 

A (A) > (B) > (C) > (D)

B (A) > (C) > (D) > (B)

C (C) > (A) > (D) > (B)

D (C) > (D) > (B) > (A)

Answer Given By Candidate: **B**

Question ID: 100178
Topic Name: Chemistry-Section A
Question:

Stearic acid and polyethylene glycol react to form which one of the following soap/s detergents ?

A Cationic detergent

B Soap

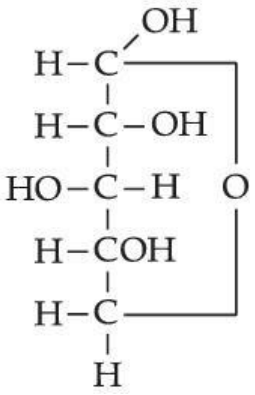
C Anionic detergent

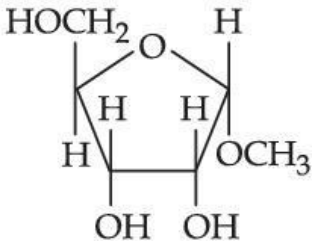
D Non-ionic detergent

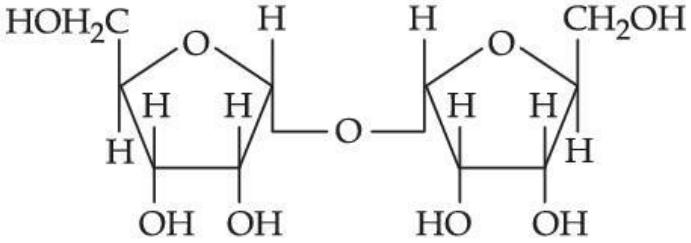
Answer Given By Candidate: **Not Attempted**

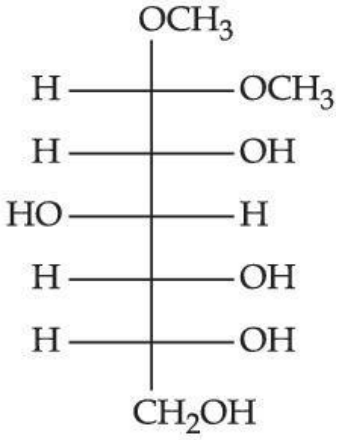
Question ID: 100179
Topic Name: Chemistry-Section A
Question: Which one of the following is a reducing sugar ?

A



B 

C 

D 

Answer Given By Candidate: **Not Attempted**

Question ID: 100180
Topic Name: Chemistry-Section A

Question:

Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Experimental reaction of CH_3Cl with aniline and anhydrous $AlCl_3$ does **not** give *o* and *p*-methylaniline.

Reason (R) : The $-NH_2$ group of aniline becomes deactivating because of salt formation with anhydrous $AlCl_3$ and hence yields *m*-methyl aniline as the product.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

A Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

B Both **(A)** and **(R)** are true but **(R)** is **not** the correct explanation of **(A)**.

C **(A)** is true, but **(R)** is false.

D **(A)** is false, but **(R)** is true.

Answer Given By Candidate:A

Question ID:100181
Topic Name: Chemistry-Section B
Question:

Chlorophyll extracted from the crushed green leaves was dissolved in water to make 2 L solution of Mg of concentration 48 ppm. The number of atoms of Mg in this solution is $x \times 10^{20}$ atoms. The value of x is _____. (Nearest Integer)

(Given : Atomic mass of Mg is 24 g mol^{-1}; $N_A = 6.02 \times 10^{23}$ mol^{-1})

Answer Given By Candidate:Not Attempted

Question ID:100182
Topic Name: Chemistry-Section B
Question:

A mixture of hydrogen and oxygen contains 40% hydrogen by mass when the pressure is 2.2 bar. The partial pressure of hydrogen is _____ bar. (Nearest Integer)

Answer Given By Candidate:Not Attempted

Question ID:100183
Topic Name: Chemistry-Section B
Question:

The wavelength of an electron and a neutron will become equal when the velocity of the electron is x times the velocity of neutron. The value of x is _____. (Nearest Integer)

(Mass of electron is 9.1×10^{-31} kg and mass of neutron is 1.6×10^{-27} kg)

Answer Given By Candidate:**Not Attempted**

Question ID:100184
Topic Name: Chemistry-Section B
Question:

2.4 g coal is burnt in a bomb calorimeter in excess of oxygen at 298 K and 1 atm pressure. The temperature of the calorimeter rises from 298 K to 300 K. The enthalpy change during the combustion of coal is $-x$ kJ mol^{-1}. The value of x is _____. (Nearest Integer)

(Given : Heat capacity of bomb calorimeter 20.0 kJ K^{-1}. Assume coal to be pure carbon)

Answer Given By Candidate:**Not Attempted**

Question ID:100185
Topic Name: Chemistry-Section B
Question:

When 800 mL of 0.5 M nitric acid is heated in a beaker, its volume is reduced to half and 11.5 g of nitric acid is evaporated. The molarity of the remaining nitric acid solution is $x \times 10^{-2}$ M. (Nearest Integer)

(Molar mass of nitric acid is 63 g mol^{-1})

Answer Given By Candidate:**Not Attempted**

Question ID:100186
Topic Name: Chemistry-Section B
Question:

At 298 K, the equilibrium constant is 2×10^{15} for the reaction :

$$Cu(s) + 2Ag^{+}(aq) \rightleftharpoons Cu^{2+}(aq) + 2Ag(s)$$

The equilibrium constant for the reaction

$$\frac{1}{2}Cu^{2+}(aq) + Ag(s) \rightleftharpoons \frac{1}{2}Cu(s) + Ag^{+}(aq)$$

is $x \times 10^{-8}$. The value of x is _____. (Nearest Integer)

Answer Given By Candidate:**Not Attempted**

Question ID:100187
Topic Name: Chemistry-Section B
Question:

The amount of charge in F (Faraday) required to obtain one mole of iron from Fe_3O_4 is _____. (Nearest Integer)

Answer Given By Candidate:**Not Attempted**

Question ID:100188
Topic Name: Chemistry-Section B

Question:

For a reaction $A \rightarrow 2B + C$ the half lives are 100 s and 50 s when the concentration of reactant A is 0.5 and 1.0 mol L^{-1} respectively. The order of the reaction is _____. (Nearest Integer)

Answer Given By Candidate:**Not Attempted**

Question ID:100189
Topic Name: Chemistry-Section B
Question:

The difference between spin only magnetic moment values of $[Co(H_2O)_6]Cl_2$ and $[Cr(H_2O)_6]Cl_3$ is _____.

Answer Given By Candidate:**0**

Question ID:100190
Topic Name: Chemistry-Section B
Question:

In the presence of sunlight, benzene reacts with Cl_2 to give product, X. The number of hydrogens in X is _____.

Answer Given By Candidate:**5**